

Youhyun Sung

Seoul, South Korea | youhyun8808@gmail.com | youhyunsung@snu.ac.kr | youhyun-sung.github.io

Research Interests

Visual perception and cognition, perceptual ambiguity and uncertainty, visual representation, perceptual decision-making, and individual differences

Research focus: how observers sample, represent, and resolve ambiguous visual information differently, leading to divergent perceptual decisions and experiences

Education

Seoul National University

2022 – 2026

M.A. in Aesthetics

Seoul, South Korea

- Completed a master's thesis in neuroaesthetics on Op Art, visual aftereffects, and their underlying perceptual and neural mechanisms.

Hanyang University

2015 – 2021

B.A. in German Language & Literature

Seoul, South Korea

- Entered with honors and graduated with honors.

University of Stuttgart

2018 – 2019

Exchange Student

Stuttgart, Germany

- Completed exchange study supported by the Baden-Wuerttemberg Stipendium.

Research Experience

Cognitive Systems Lab, Korea University

Nov. 2025 – Present

Research Assistant

Seoul, South Korea

- Working under Prof. Christian Wallraven in Brain and Cognitive Engineering.
- Investigating how observers resolve ambiguous visual information differently, using Op Art and matched visual stimuli as controlled experimental domains.
- Designing behavioral paradigms examining perceptual disagreement, confidence, viewing behavior, and aesthetic judgments under visual uncertainty.
- Developing eye-tracking pilot studies to examine how visual sampling and gaze dynamics relate to subjective perceptual experience.
- Preparing manuscripts and conference work for *Art & Perception*, VSAC, ECVP, and related outlets.

Publications & Manuscripts

Sung, Y., Sung, M., Carbon, C., & Wallraven, C.

In preparation

More Ambiguous, Not More Preferred: Spatial Ambiguity and Aesthetic Preference in Op Art.

Sung, Y.

Preprint available

Op Art and Its Visual Effect: Focusing on the Neural Mechanisms of Aftereffects.

- Master's thesis, Seoul National University. OSF: 10.17605/OSF.IO/M32EV.

Conference Presentations

Visual Science of Art Conference (VSAC)

2026

Does Spatial Ambiguity Contribute to Aesthetic Preference? Evidence from Op Art and Matched Control Stimuli.

- Selected for oral presentation after original poster submission.

48th European Conference on Visual Perception (ECVP)

2026

Art wins: Higher Valuation Responses to Op Art Compared to Matched Optical Illusions.

Research Projects

Seeing Ambiguity

Individual differences in perceptual ambiguity and visual decision-making

- Examines why observers reach different perceptual judgments from the same ambiguous visual information, using behavioral disagreement, confidence, and viewing dynamics as complementary measures.

Tracing the Invisible

Eye-tracking pilot study on afterimages and visual perception

- Examines how gaze behavior tracks the emergence and fading of subjective afterimages, linking visual sampling dynamics to internally generated perceptual experience.

Art? Illusion?

Op Art, optical illusions, and aesthetic valuation

- Examines willingness-to-pay judgments for Op Art images and matched optical illusion images.

Work Experience

Alto

2022 – 2025

Freelance Event & Exhibition Staff

Seoul, South Korea

- Supported Samsung IFA Berlin 2025, K-Science UK at the Korean Cultural Centre UK, Hyundai Motorstudio Heritage Exhibition Launching Show, Nike Jordan, and Nike Global projects.

Cheil Worldwide

2019 – 2026

Freelance Event & Brand Experience Staff

Seoul, South Korea

- Supported Samsung ILP New York 2025, Samsung ILP New York 2024, Samsung IFA Berlin 2023, and Samsung IFA Berlin 2019.

Frieze Seoul

2023 – 2025

Freelancer

Seoul, South Korea

- Supported event operations in an international contemporary art fair environment.

10th World Congress of Constitutional Law

Feb. 2018 – Jun. 2018

Part-time Operation & Planning Staff, Sungkyunkwan University

Seoul, South Korea

- Assisted program operations and planning for an international academic conference.

Teaching & Service

Homo Aestheticus Seminar

2026

Organizer, Korea University

Seoul, South Korea

- Organized a seminar featuring visiting professor Claus-Christian Carbon.

Seoul National University

2025

Teaching Assistant, Art and Science

Seoul, South Korea

- Supported course delivery and instruction across one academic year.

Awards & Scholarships

Seoul National University Mentoring Scholarship

2022 – 2023

Recipient

Seoul, South Korea

- Awarded mentoring scholarship during graduate study.

Baden-Wurttemberg Stipendium

2018 – 2019

Recipient

Germany

- Scholarship support of approximately EUR 7,000 for exchange study at the University of Stuttgart.

Academic Excellence Scholarship, Hanyang University

2016 – 2017

Recipient

Seoul, South Korea

- Received merit-based academic scholarship.

Languages & Certificates

Languages

Korean (Native), English (Fluent, C1), German (Fluent, C1), French (Basic, A2–B1)

Selected Certificates

IELTS Overall 8 (LC 8.5, RC 8.5, WR 7.5, SP 7.5)

TOEFL iBT 107 (LC 26, RC 26, WR 28, SP 27); Goethe Zertifikat Deutsch C1

Research & Technical Skills

Research

Psychophysics, behavioral experiment design, perceptual decision-making paradigms, visual stimulus control, eye-tracking, data cleaning, and statistical analysis

Technical

JavaScript, HTML/CSS, Python, pandas, EEG, eye-tracking (Tobii Pro), GitHub, OSF, LaTeX/Overleaf